



ABACUS GLOBAL

ESS

A SMART SOLUTION FOR
PREDICTING ENVIRONMENTAL
ANOMALIES.



Presented By:

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INTRODUCTION

ENVIRONMENT AND COMFORT

Personal comfort has been a long sought after goal for humanity.

Generally, if local conditions are just right, everyone in the area feels satisfied. Yet, there are times when local opinion does not match official data at all.

This means that regions with seemingly perfect environments find themselves suddenly abandoned by tourists.

Fast, dynamic responses

are necessary for tourist reliant cities and industries. To match these demands, it is necessary to harness the power of digital twins and AI to fill the gap when regular data cannot.



SOLUTION

The ESS system simulates environmental data in real-time, all while gathering subjective local views as well.

This is coupled with a robust predictive system that uses past data to simulate what locals would be *expected* to feel.

This immediate detection enables operators to act quickly, addressing issues before they escalate into broader dissatisfaction or tourist decline.

- ✓ **Dealing with Rapid Environmental Changes**
- ✓ **Detecting Dissatisfaction Early**
- ✓ **Increasing Tourist Loyalty and Satisfaction**



ARCHITECTURAL DIAGRAM



DATA SOURCING

ESS sources live temperature data from OpenWeather, a UK-based company that provides live weather data.

Live air quality data is sourced from aqicn.org.

Live local comfort views are obtained using a proprietary Flutter based Android app

Due to the unavailability of free historical data, past local views & historical AQI data for Lahore, Islamabad and Karachi have been simulated.

Further Data Usage



Sonamarg, Kashmir - A tale of deceptive weather.

“

A once thriving ski resort has found itself abandoned. Despite weather conditions appearing perfect on the surface, climate change has caused snowfall patterns to become erratic, making it unsuitable for winter sport.

Here, digital twin systems such as ESS would have allowed operators to observe subtle changes, by comparing predictions to current local comfort. Thus making further investigations and changes in investment strategy possible before it was too late.

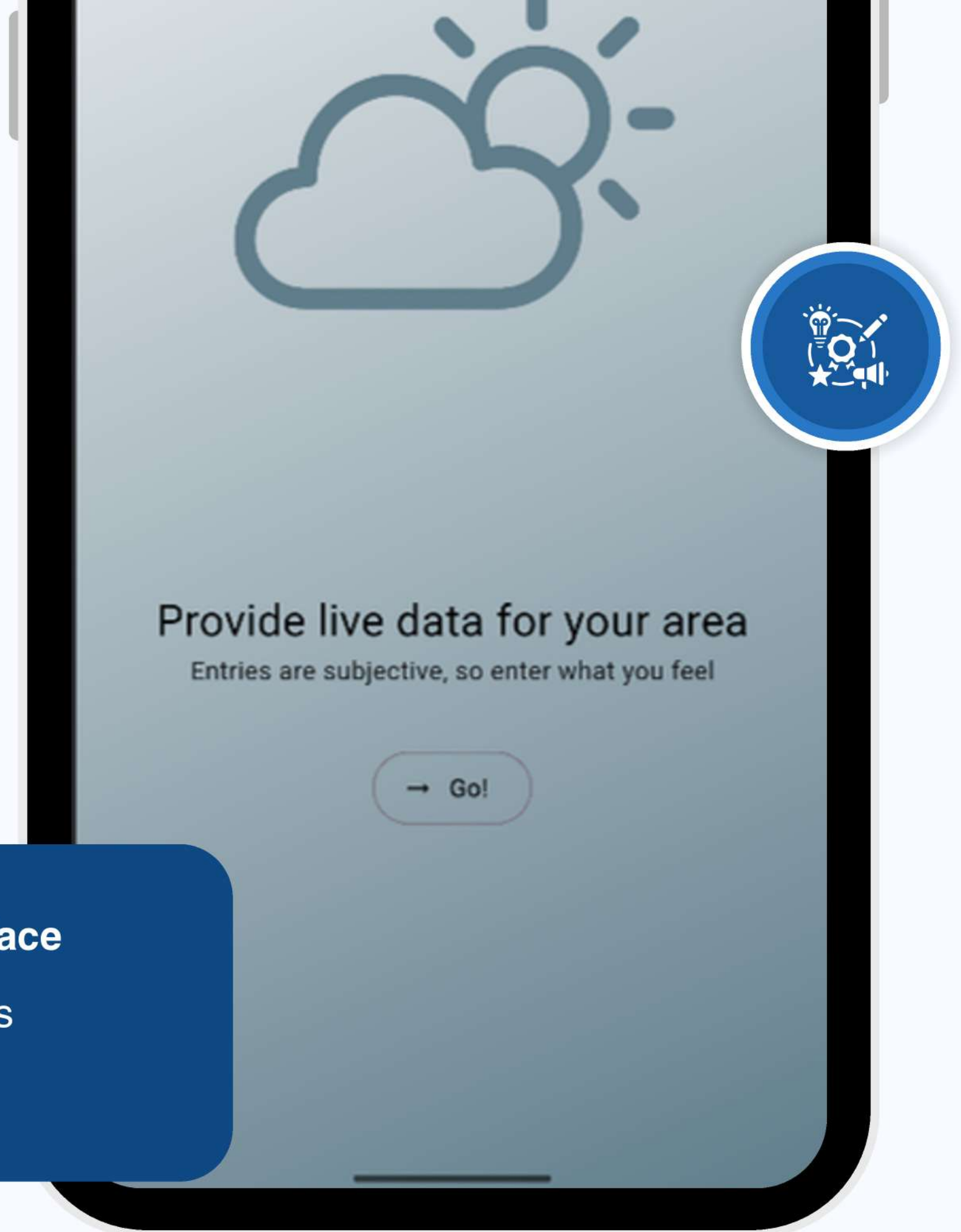
FRONT-END: FLUTTER

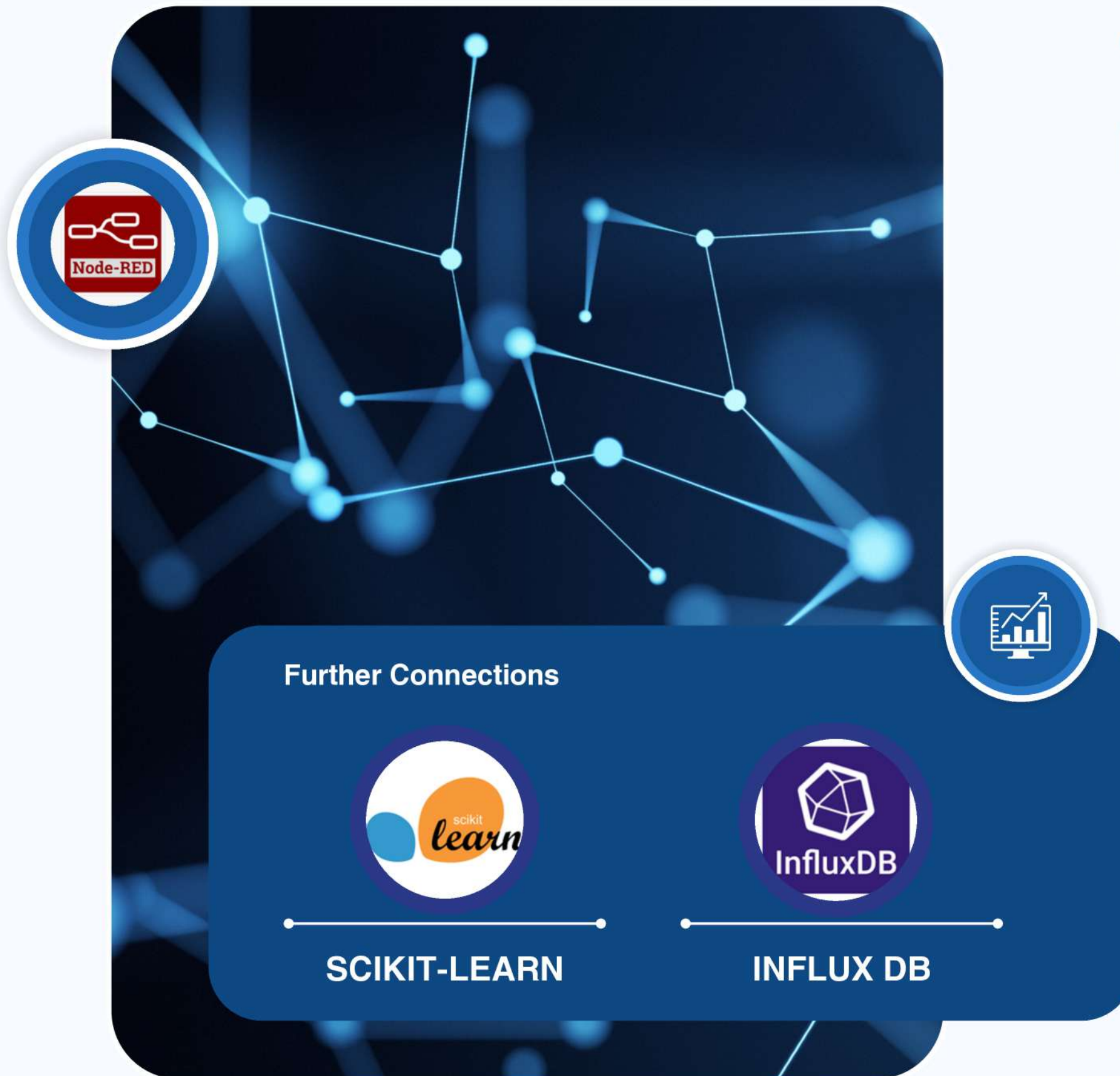
Flutter provides an easily maintained framework to build and maintain apps, proving to be a robust solution for a user-facing app for locals to enter current information about their comfort.

This data is sent directly to Influx via Node-RED.



- ✓ **Provides a simple, user friendly interface**
- ✓ Supported by all modern Android devices
- ✓ **Connects easily to Node-RED**





THE BACKBONE: NODE-RED

Node-RED is a js-based visual editor that allows for hardware, APIs, and online services to be configured into event-driven flows.

ESS uses Node-RED to connect its data sources to their endpoints.

Why Node-RED?

Provides a robust, open source architecture

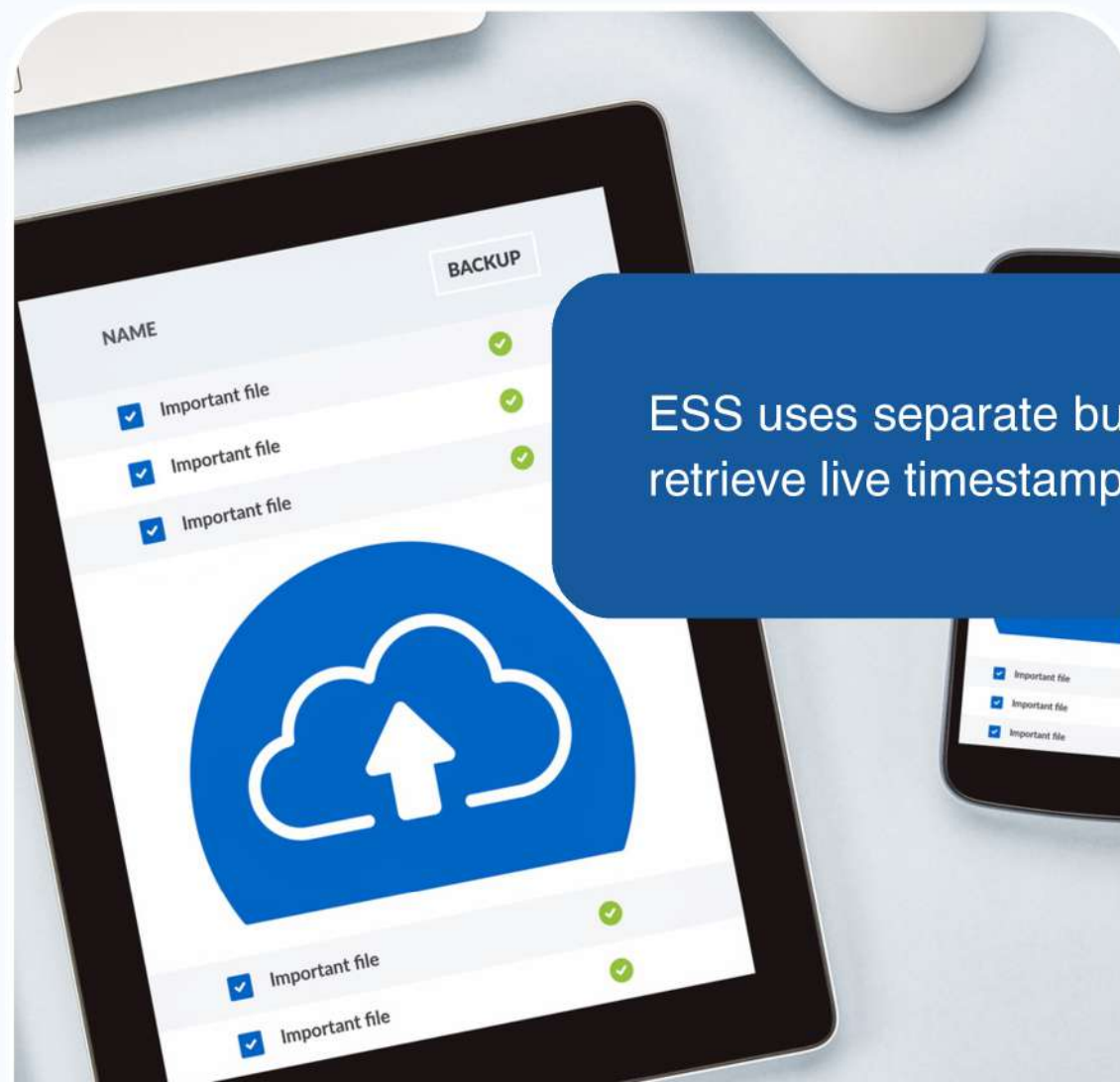


Supports a wide range of tools, databases and frameworks



THE DATABASE: INFLUX-DB

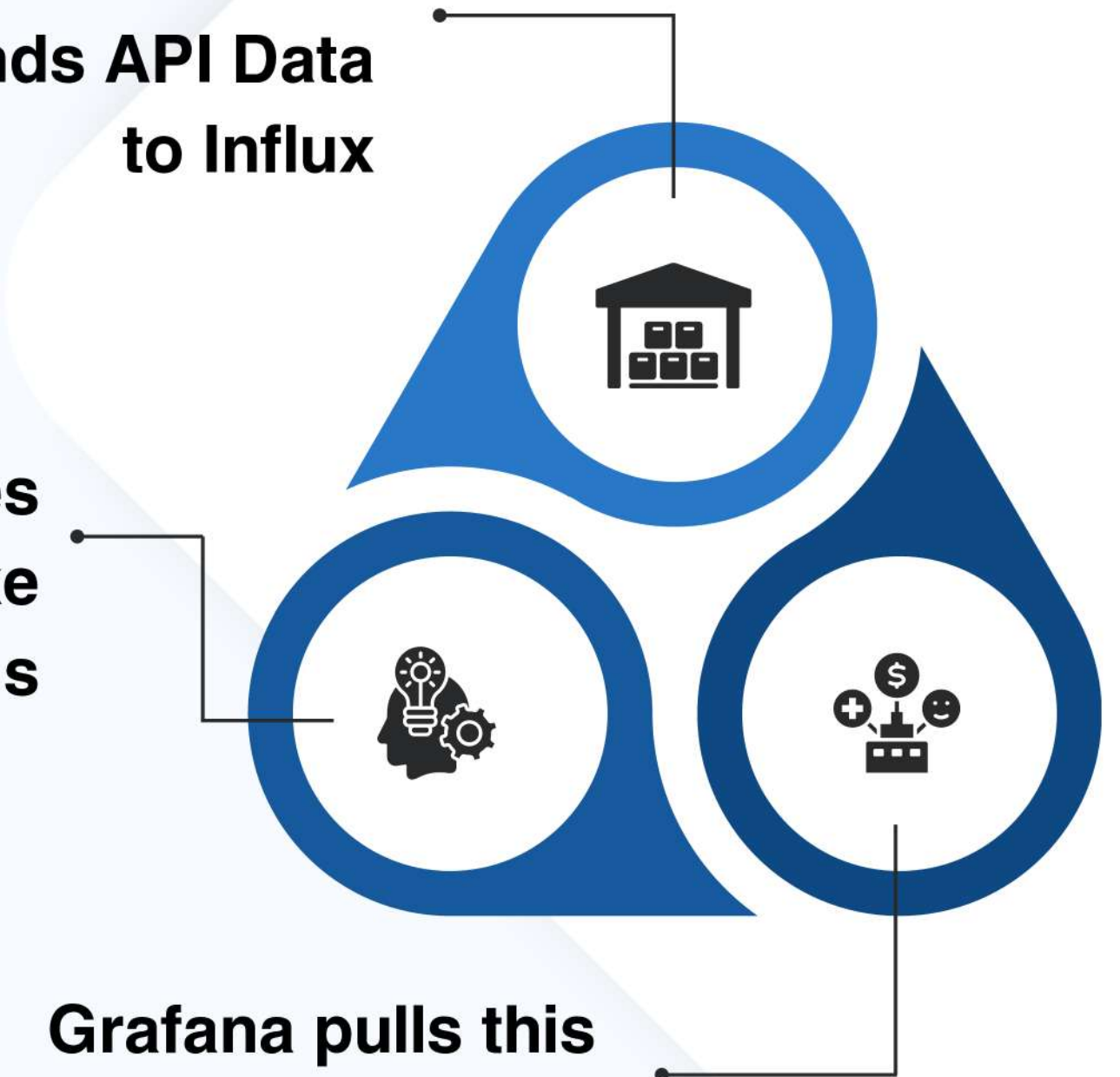
InfluxDB is an open-source time series database developed by InfluxData, optimized for the efficient storage, querying, and analysis of timestamped data. In this case, these are temperature, US API and local satisfaction.



ESS uses separate buckets for each city, to store and retrieve live timestamped data in real-time

Scikit-Learn uses this to make predictions

Node-RED sends API Data to Influx



Grafana pulls this data to visualize it

PREDICTIONS: SCIKIT-LEARN

Scikit-Learn is a Python-based library that provides numerous easily configurable statistical and machine learning algorithms.

ESS uses a random forest model to make informed predictions on current local comfort.



Leveraging Scikit-Learn for state-of-the-art algorithms

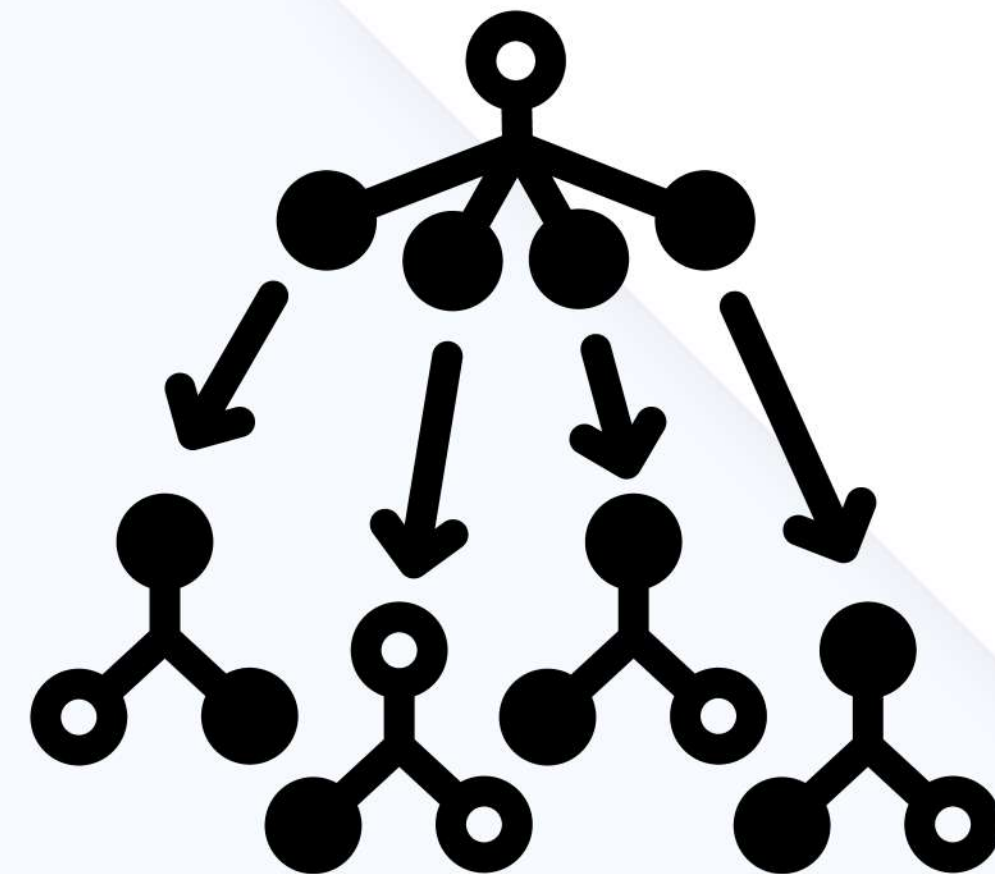
- ✓ ADAPTABLE, EASILY TRAINED MODELS
- ✓ EXTENSIVE DOCUMENTATION & SUPPORT
- ✓ STRONG INTEGRATION WITH PYTHON





RANDOM FOREST MODELLING

The Random Forest algorithm builds a forest of decision trees, training on a random bootstrap sample of data and considering only a random subset of features at each split. It aggregates their outputs via majority vote (for classification) or average (for regression).

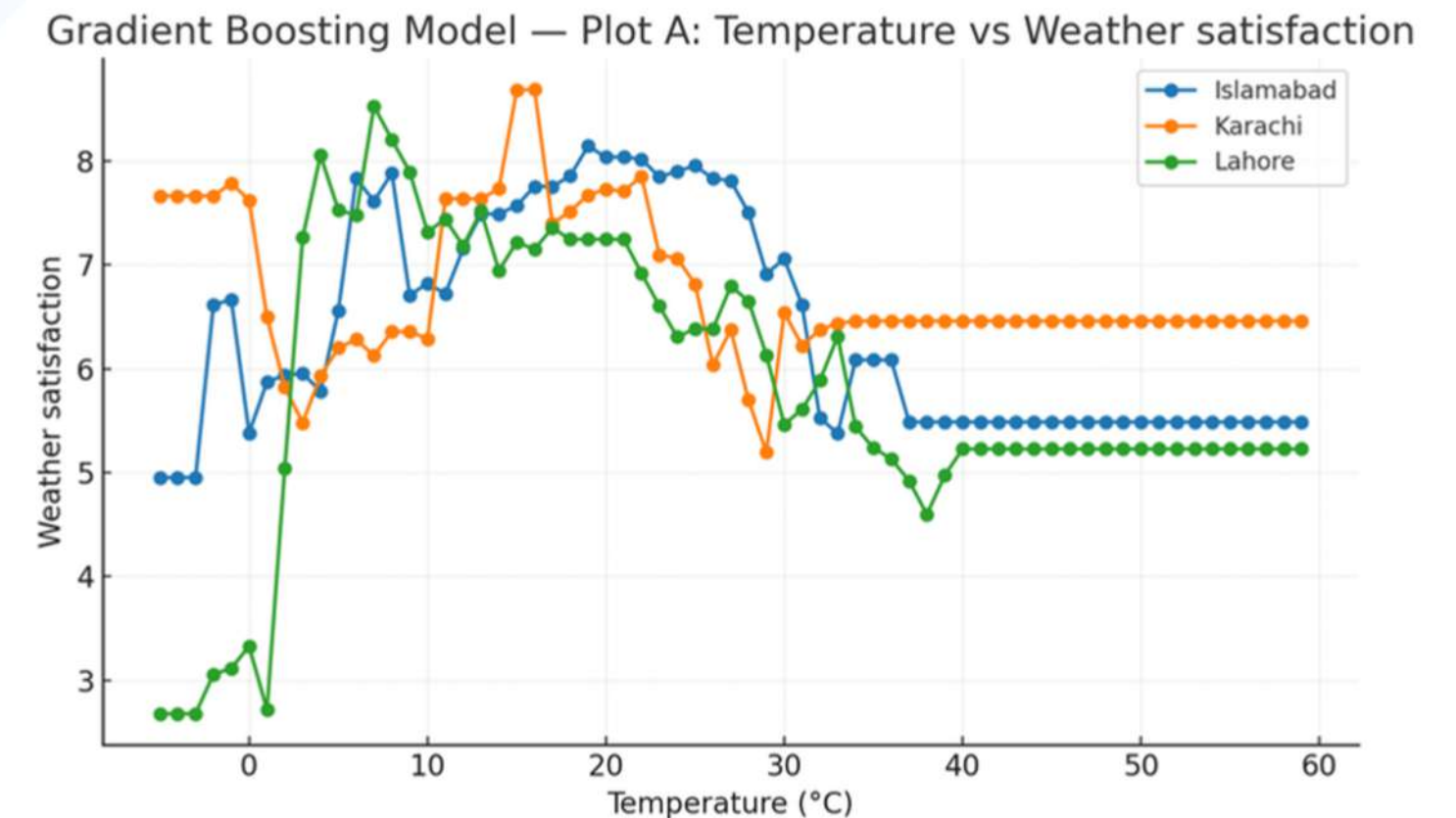
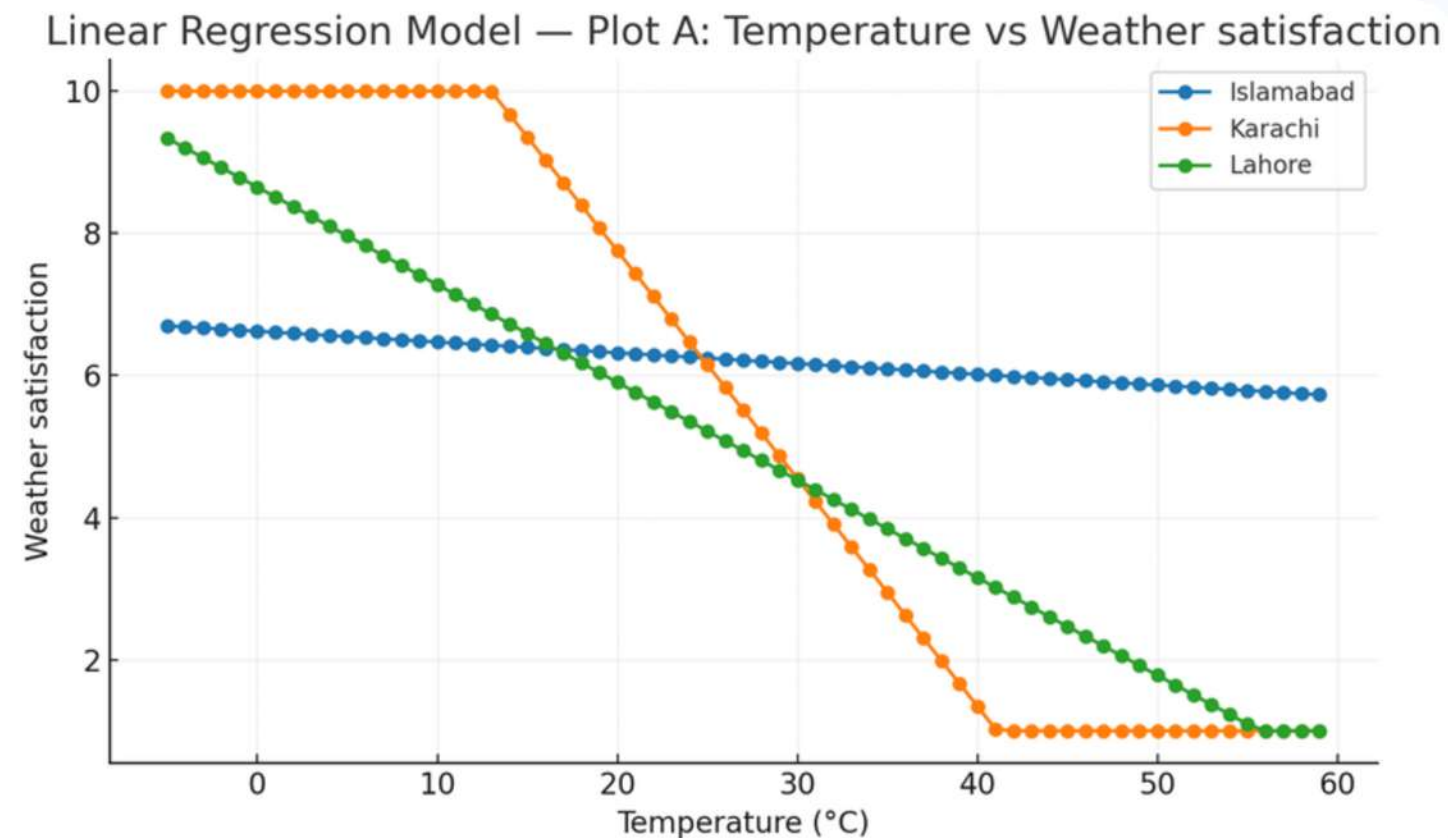
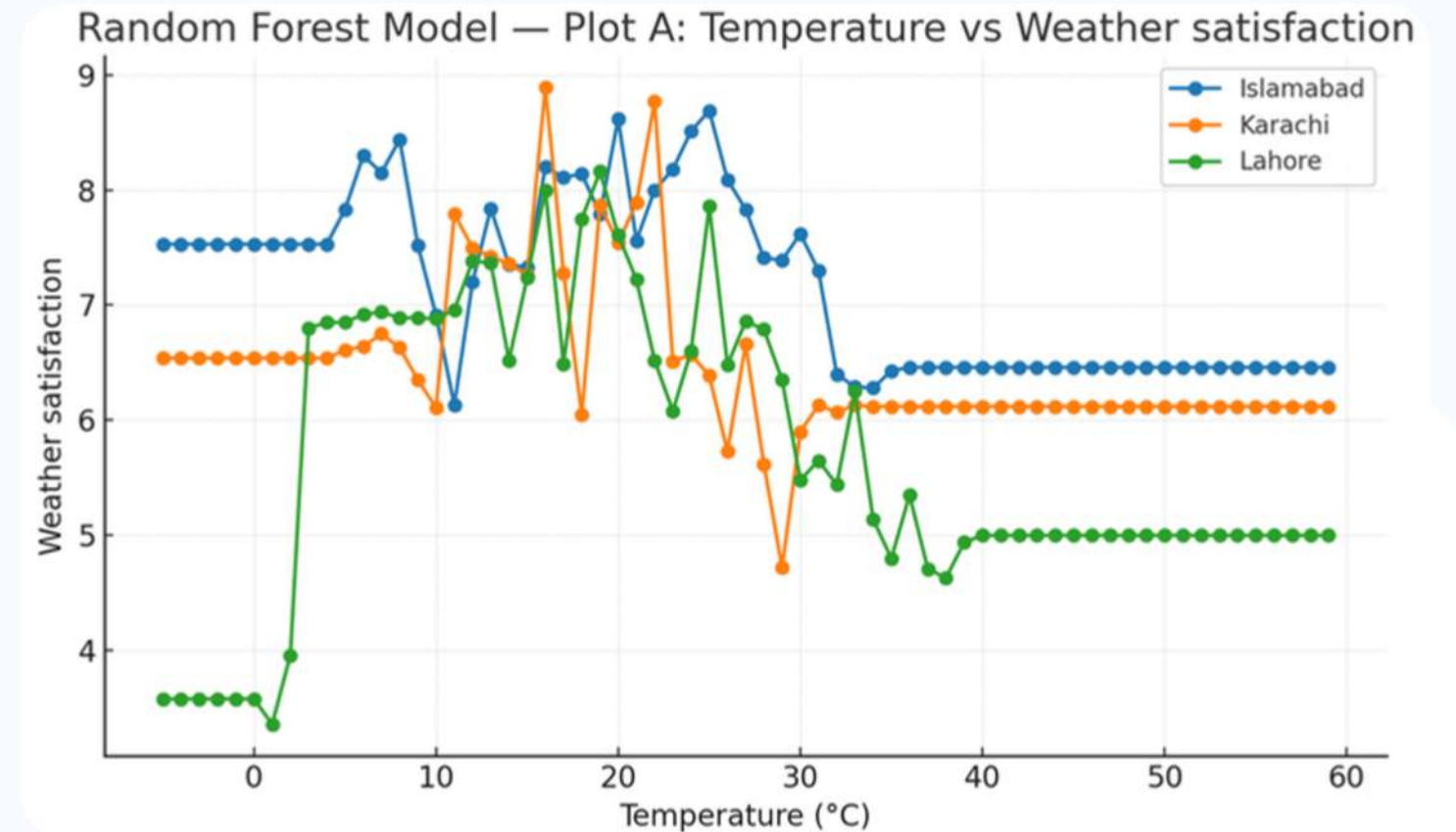


TESTS: TEMPERATURE

The random forest model was chosen after tests with numerous other algorithms, those being linear regression, and gradient boosting.

The results are shown, with the predicted satisfaction value on the y axis, and the input temperature value on the x-axis.

The tests were run for a range of temperatures, from -5 to 60 degrees Celsius.

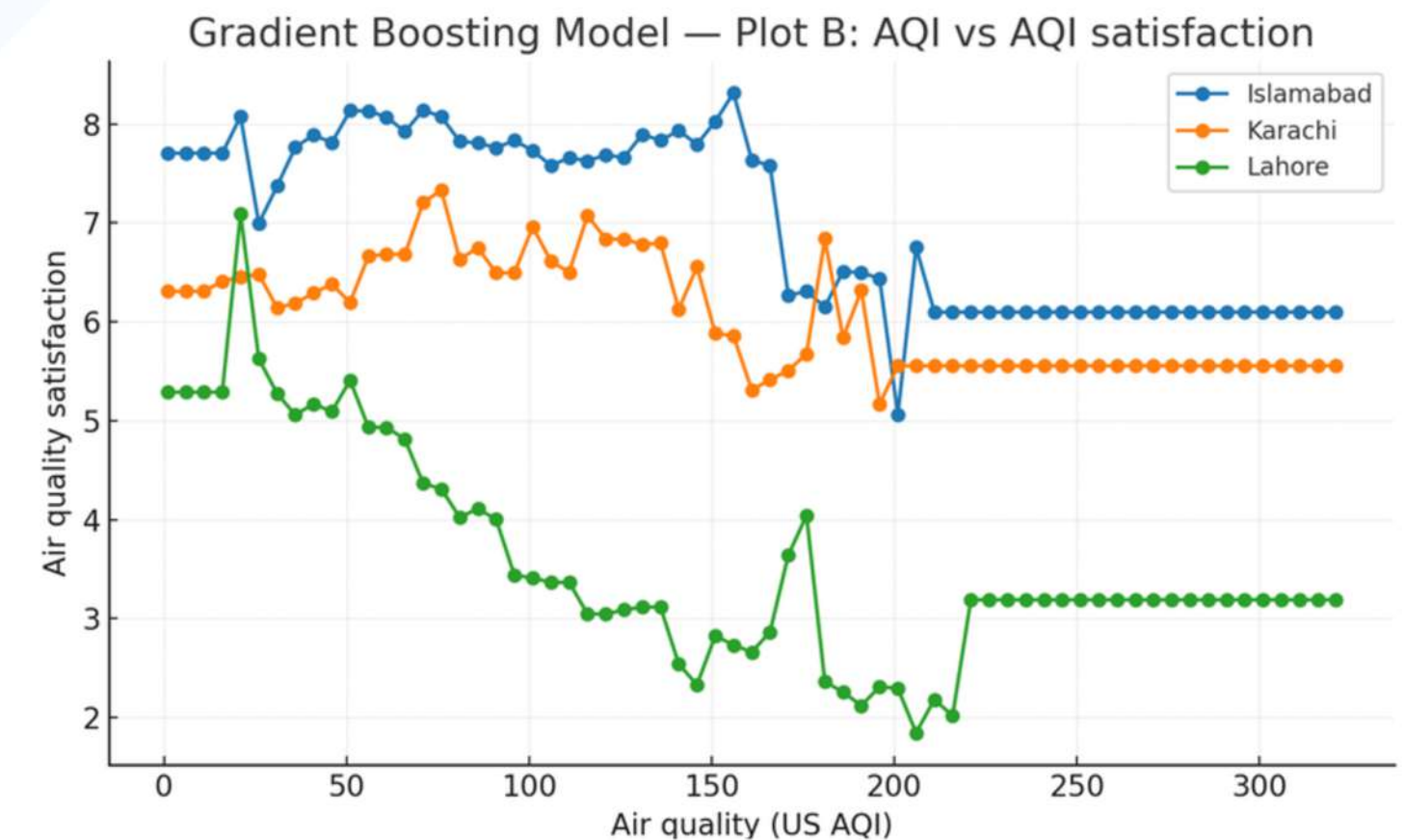
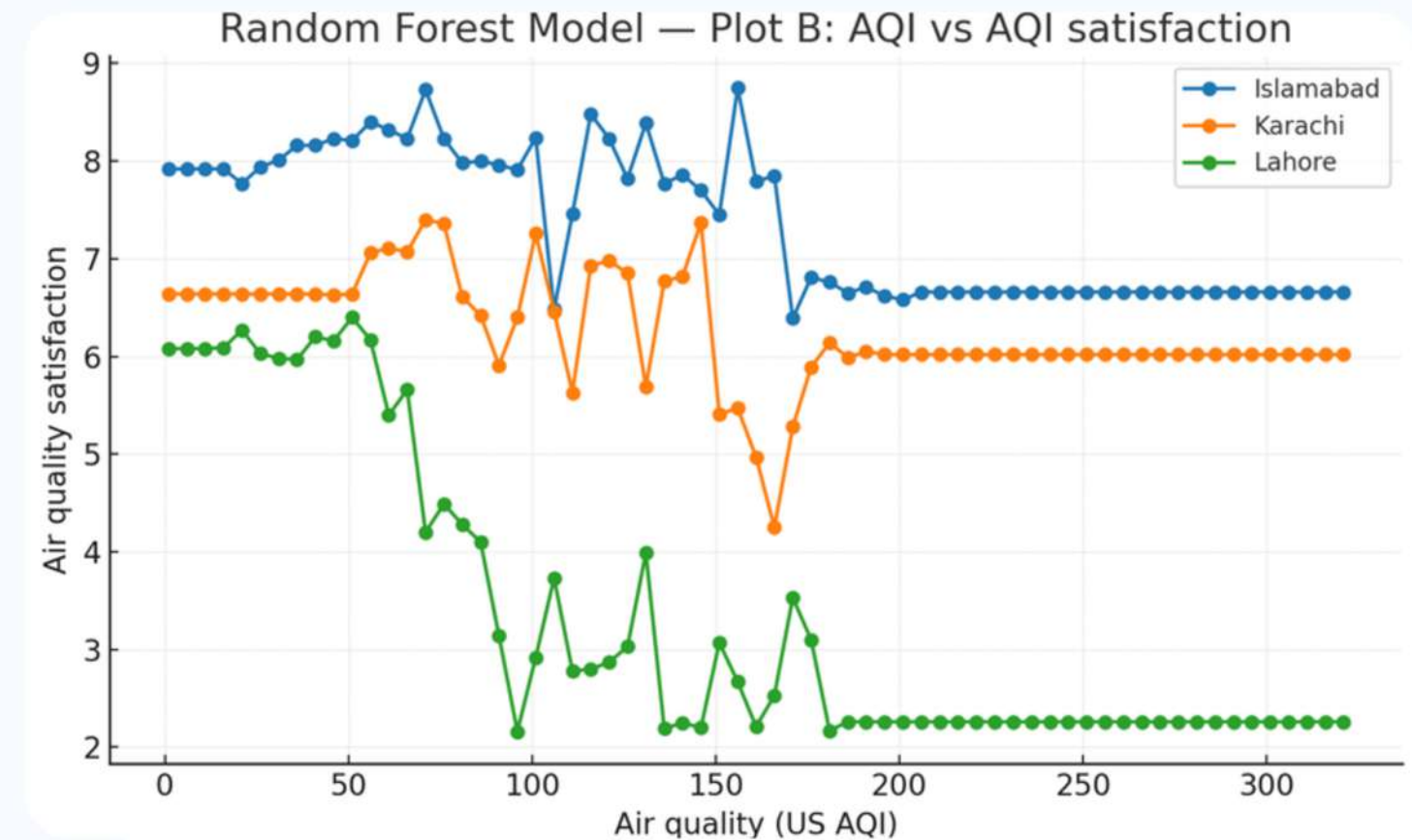
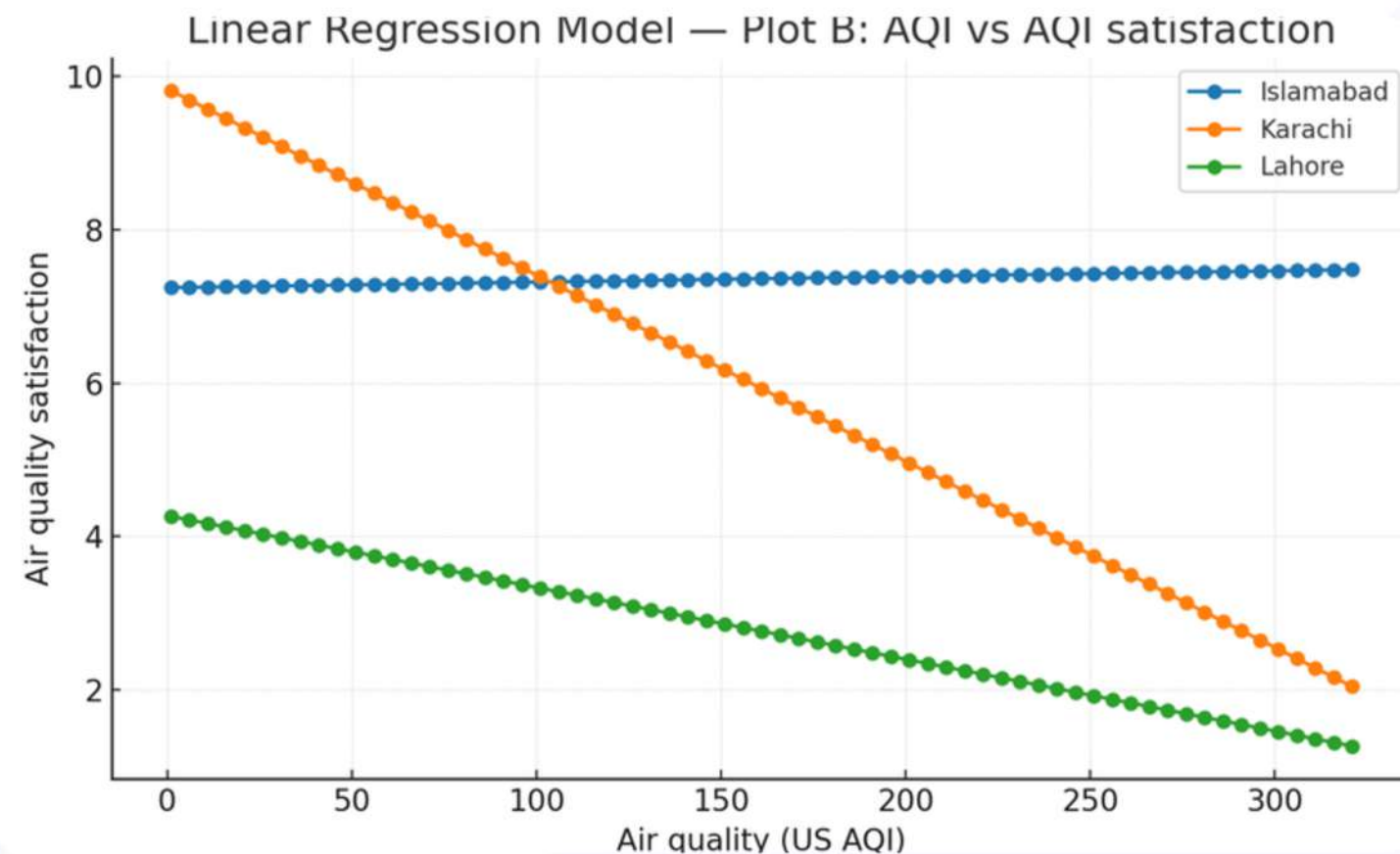


TESTS: US AQI

The random forest model was chosen after tests with numerous other algorithms, those being linear regression, and gradient boosting.

The results are shown, with the predicted satisfaction value on the y axis, and the input temperature value on the x-axis.

The tests were run for a range of AQI levels, from 0 to 350.

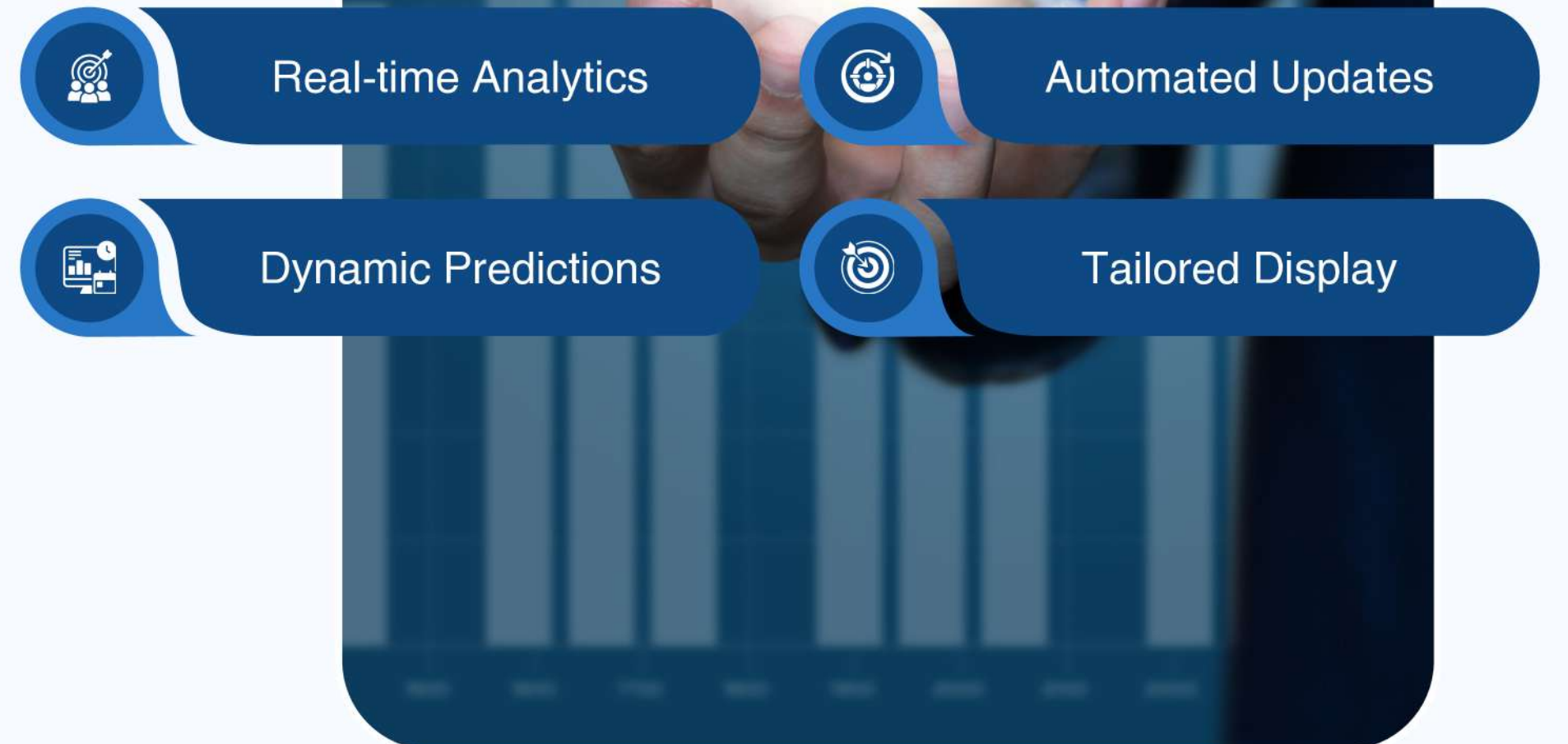
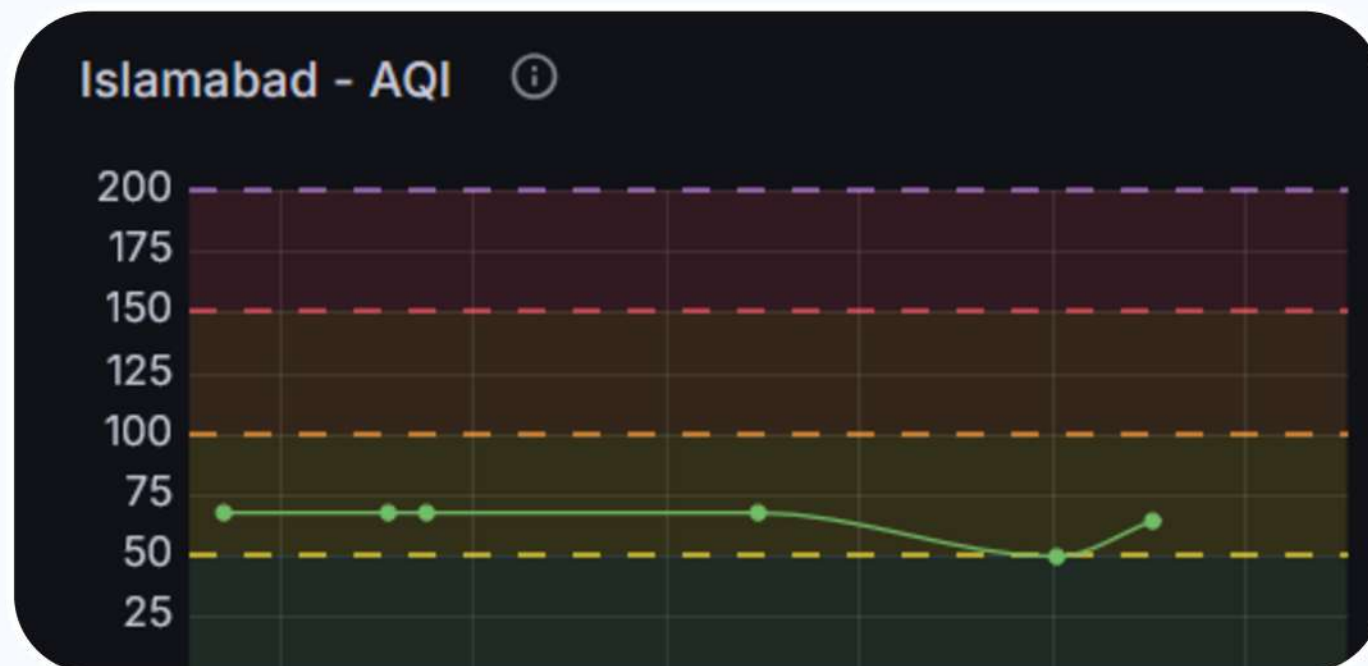


VISUALIZATION: GRAFANA

Grafana is an open-source, multi-platform web application for querying and visualizing data, all via customizable dashboards.

It supports a direct connection with InfluxDB, allowing for a robust, streamlined connection.

The ESS dashboard allows operators to monitor five distinct metrics across three Pakistani cities, Lahore, Islamabad and Karachi.



DEMONSTRATION

EVALUATION

CONCLUSION

AND RECOMMENDATIONS

ESS is perfect for data-driven decisions in the tourist industry, providing operators with analytics that allow them to respond as soon as tourists begin to feel discomfort. This can potentially vast amounts of



THANK YOU

FOR YOUR ATTENTION
AND PARTICIPATION

